Code No: 136AG **R16**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech III Year II Semester Examinations, July - 2023
ARTIFICIAL INTELLIGENCE
(Computer Science and Engineering)

Time: 3 Hours **Max. Marks: 75**

Note: i) Question paper consists of Part A, Part B.
ii) Part A is compulsory, which carries 25 marks. In Part A, Answer all questions.
iii) In Part B, Answer any one question from each unit. Each question carries 10 marks and may have a, b as sub questions.

PART – A

(25 Marks)

1.a)	List the fields that form the basis for AI.	[2]
b)	Define alpha and beta values in a game tree.	[3]
c)	How do frames differ from semantic nets?	[2]
d)	Differentiate prepositional and predicate logic.	[3]
e)	List the characteristic features of an expert system.	[2]
f)	What is Knowledge Acquisition?	[3]
g)	Define training and testing data.	[2]
h)	What are the types of problems in which Artificial Neural Network can be applied?	[3]
i)	What is n-gram language model?	[2]
j)	What are the objectives of NLP?	[3]

PART – B

(50 Marks)

2. What is A* search? Explain various stages of A* search with an example. [10]

OR

3. Summarize about the following:
a) Bounded Look-ahead Strategy
b) Alpha-Beta Pruning. [5+5]

4. Consider the following facts and represent them in predicate form:
F1. There are 500 employees in ABC company.
F2. Employees earning more than Rs. 5000 pay tax.
F3. John is a manager in ABC company.
F4. Manager earns Rs. 10,000.
Convert the facts in predicate form to clauses and then prove by resolution: "John pays tax". [10]

OR

5. Explain resolution in predicate logic with suitable example. [10]

used paper July/Aug-2023

6. Describe in detail about Expert System Architecture. [10]

OR

7.a) Differentiate between Expert Systems and Traditional Systems.

b) What are the application of Expert Systems? Explain. [5+5]

8. State the mathematical formulation of the SVM problem. Give an outline of the method for solving the problem. [10]

OR

9. Show the final result of hierarchical clustering with complete link by drawing a dendrogram. [10]

	A	B	C	D	E	F
A	**0**					
B	**0.12**	**0**				
C	**0.51**	**0.25**	**0**			
D	**0.84**	**0.16**	**0.14**	**0**		
E	**0.28**	**0.77**	**0.70**	**0.45**	**0**	
F	**0.34**	**0.61**	**0.93**	**0.2**	**0.7**	**0**

10. Give the five logical connectives used to construct complex sentences and give the formal grammar of propositional logic. [10]

OR

11. Explain in detail the following:
a) Semantic Analysis
b) Universal Networking Knowledge. [5+5]

---ooOoo---

used paper July/Aug-2023